HAIYI MEI

Research Fellow | Synthetic Data Engineer | Backend Engineer | Game Developer

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TECH STACK

- Python: 6+ years
- C++: 7+ years
- PyTorch: 6+ years
- Unreal Engine: 4+ years
- Blender: 3+ years
- Git
- Docker
- CI/CD

ACQUAINTED

- Game Engine
- PBR
- Synthetic Data
- SMPLX
- SaaS
- NeRF
- Diffusion
- Differentiable Rendering
- Music (Piano/Guitar/Theory)

EDUCATION

Master's Degree

Shandong University

2019 – 2022

Bachelor's Degree

NJUST

2015 – 2019

- Head of the Symphony Orchestra of School

LANGUAGES

Chineses:	**Native**
English:	**C1**
Japanese:	**A1**

ABOUT ME

I'm a committed algorithm researcher and backend engineer with 3 years of industry experience. Specializing in 3D synthetic data generation for computer vision tasks, I contributed to 5 papers in top-tier conferences. I possess expertise in game engines, computer graphics, and machine learning. I'm into engine tools development, backend development, and built open-source projects and scalable robust systems.

EXPERIENCE

Mid-Level Research Fellow | SenseTime

2021 – Present

- Spearheaded the development of XRFeitoria, a rendering toolbox that streamlines synthetic data generation. This tool simplifies the construction of rendering pipelines in diverse areas, enabling contribution to publishing 5 papers within 1 year.
- Mainrole in designing SynBody, a large-scale synthetic dataset with layered human models.
  - Participated in designing SMPL-XL, which enriches SMPL-X in hair, garments, accessories, and textures.
  - Built a System of synthetic data generation as a Software-as-a-Service (SaaS) platform, including 1) layered human creation, 2) motion retargeting, 3) scene composition, 4) rendering pipeline, and 5) flow control.
  - Published a paper presented at ICCV 2023 as co-first author.
- Proficient in crafting advanced technology demonstration videos.

Research Intern | NLPR, CASIA

2018 – 2021

- Started to work on synthetic data generation for computer vision tasks.
  - Patent filed with CNIPA: CN113011334A.
  - One paper related to synthetic data.
- Had the experience of reproducing the SOTA methods in video captioning. Code

OPEN-SOURCE PROJECTS

XRFeitoria | Synthetic Data Rendering Toolbox

- Control over engine (UE/Blender) through RPC using system python.
- Render photorealistic images with ground-truth annotations.
- Manage assets/cameras, including import, place, export, and delete.
- CLI tools to render images from a mesh file.
- Streamline building rendering pipelines across various domains, effectively implemented in over 6 projects.

AWARDS

Patents
Hold 4 patents filed with CNIPA
CN108453742B, CN113011334A, CN114429436A, CN115272548A

Music Award – 2017
First Place Award, National University Piano Competition
Chinese Golden Bell Award for Music